

GE261004 DESIGN THINKING

Unit IV – Prototyping and Testing · Presentation 4 of 6

Rapid Prototyping

From sketch to solid · what CAD adds · FDM, SLA and SLS · laser cutting and CNC · digital mock-ups · what rapid prototyping does not fix

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In this presentation

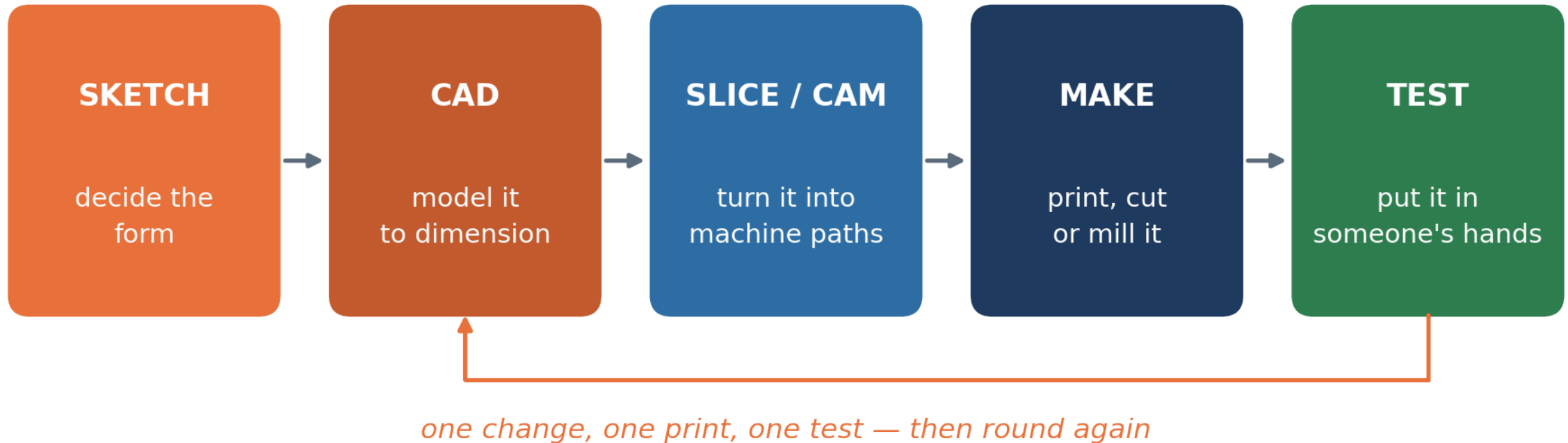
- From sketch to solid
- What CAD adds
- Three additive processes
- The processes compared
- Subtractive and forming methods
- Choosing a method
- Digital mock-ups for screens
- What it does not fix

1

From sketch to solid

The pipeline, and what each step costs you

The rapid prototyping pipeline



Five steps and one loop – the loop is where the value is.

What CAD adds

- Dimensions: a part that will actually fit the part next to it
- Repeatability: change one number and remake it identically
- A file you can send to a machine, a supplier or a classmate
- Analysis: mass, interference, clearance and stress, before anything is cut

KEY POINT CAD is where a shape becomes a part. It is also where teams start polishing instead of deciding – sketch first.

2

Making the part

Adding, removing and forming material

Three additive processes

FDM

a heated nozzle lays
molten plastic in layers

cheapest, strong enough
for most class work

SLA

a light source cures
liquid resin layer by layer

finest detail and the
smoothest surface

SLS

a laser fuses powder,
which supports itself

no supports needed;
complex shapes

Chuck Hull patented stereolithography in 1986; Scott Crump developed FDM in 1988.

The processes compared

	FDM	SLA	SLS
Builds with	Melted filament	Cured resin	Fused powder
Detail	Visible layers	Very fine	Fine, matte
Supports	Needed	Needed	Not needed
Cost	Lowest	Middle	Highest
Best for	Fit and form	Appearance	Complex shapes

Subtractive and forming methods

- **Laser cutting** – flat parts from acrylic, ply or card; fastest route to a chassis or an enclosure
- **CNC milling** – real material, real strength, real tolerances; slower and needs fixturing
- **Vacuum forming** – thin shells over a printed mould; good for covers and housings
- **Hand tools** – still the fastest way to change a part that is already in front of you

KEY POINT Flat parts and a laser cutter will get you a testable enclosure in an afternoon. A printed one takes overnight.

Choosing a method

- What question are you answering – does it fit, does it look right, or does it hold load?
- How many do you need, and how soon do you need them?
- Does it have to be the real material, or only the real shape?
- What can you get onto a machine before the next test session?

KEY POINT The right process is the one that puts something in a user's hands before your next class, not the one with the best specification.

Digital mock-ups for screens

What they are good for

- Testing a flow before any code
- Comparing two layouts
- Sharing with a remote user
- Changing in minutes

What they hide

- How slow it really is
- What happens when it fails
- Typing on a real phone
- Use in sunlight, or one-handed

KEY POINT A click-through only ever tests the path you thought of. Paper lets the user go somewhere you did not draw.

What rapid prototyping does not fix

- A printed part of the wrong idea is still the wrong idea, in better plastic
- Machine time tempts teams to build one careful thing instead of three rough ones
- Fidelity invites politeness – people criticise a printed part far less freely
- The file is not the design; the design is what happens when somebody uses it

KEY POINT Rapid prototyping makes the making fast. It does nothing at all about choosing the right thing to make.

In class: take one part to CAD

- Pick the single part of your prototype whose dimensions actually matter
- Model only that part – not the assembly, not the housing
- Decide the process, and say in one sentence why that process
- Everything else stays cardboard until the test says otherwise

ACTIVITY Forty minutes. Each team submits one CAD file and one sentence justifying the process it chose.

THINK ABOUT IT

Which part of your prototype genuinely needs to be printed – and which one are you printing because it is more fun?

- What would cardboard fail to tell you about that part?
- How long is the print, and what will you do while it runs?
- If the print fails overnight, what is your plan for the test?
- Could you test the same question with something already in the room?

Summary

- Sketch, CAD, slice, make, test
– and the loop back to CAD
- CAD adds dimension, repeatability and a file a machine can read
- FDM is cheapest, SLA finest, SLS needs no supports
- Laser cutting gives a testable enclosure in an afternoon
- Choose the process by the question and the deadline, not the specification
- Rapid making does nothing about choosing the right thing to make

Next: Testing methods and user feedback

- What a test is actually for
- Thirty minutes with one user
- How many users you need
- Capturing feedback so a team can act on it